CSS



What is CSS?

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files

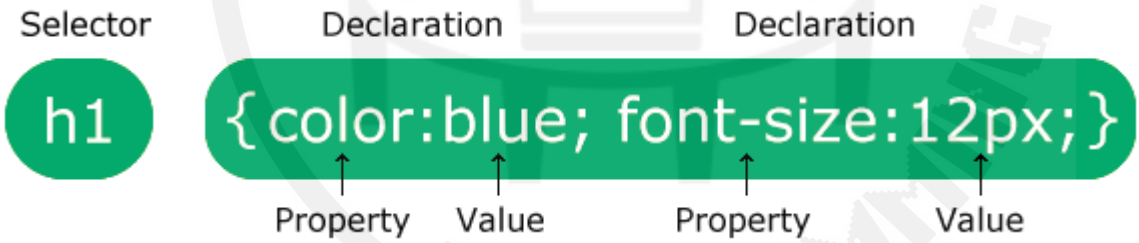
Why we need CSS?

.To style the webpage we need CSS.



CSS Syntax:

- **CSS syntax consists of a selector and declaration block**



Three Ways to Insert CSS:

There are three ways of inserting a style sheet:

- External CSS
- Internal CSS

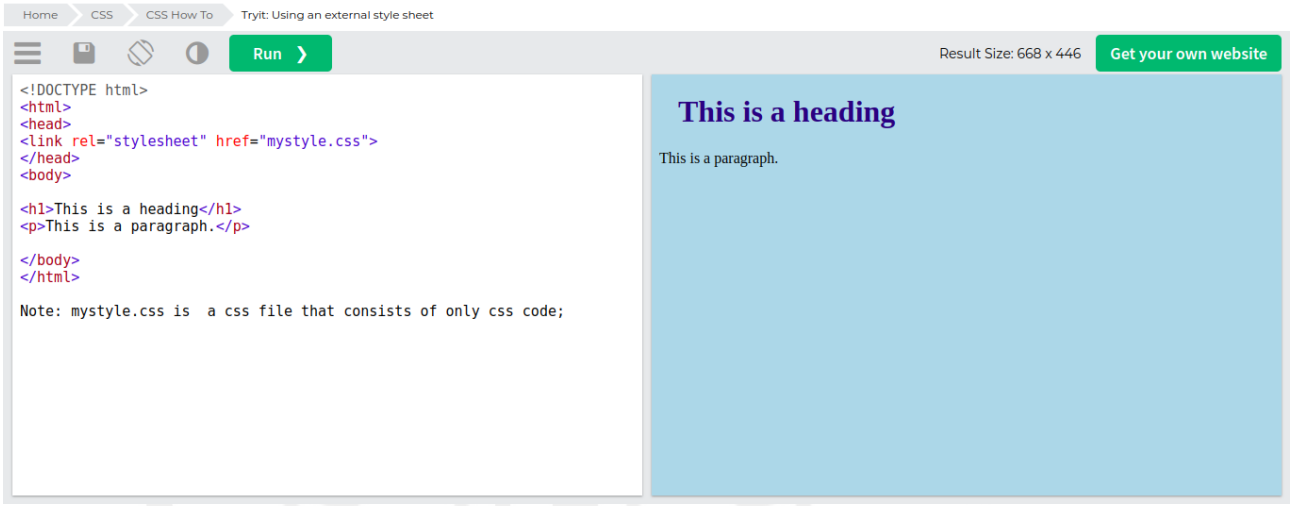
- Inline CSS

1.External CSS

With an external style sheet, you can change the look of an entire website by changing just one file!

Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section.

SAMPLE CODE AND OUTPUT:

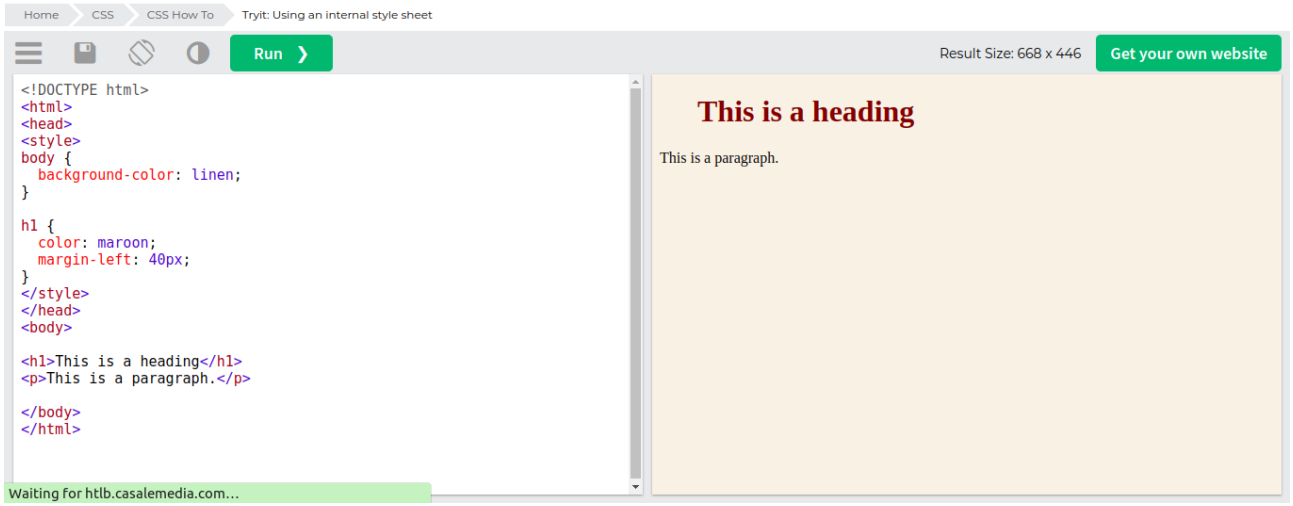


2.Internal CSS

An internal style sheet may be used if one single HTML page has a unique style.

The internal style is defined inside the <style> element, inside the head section.

SAMPLE CODE AND OUTPUT:

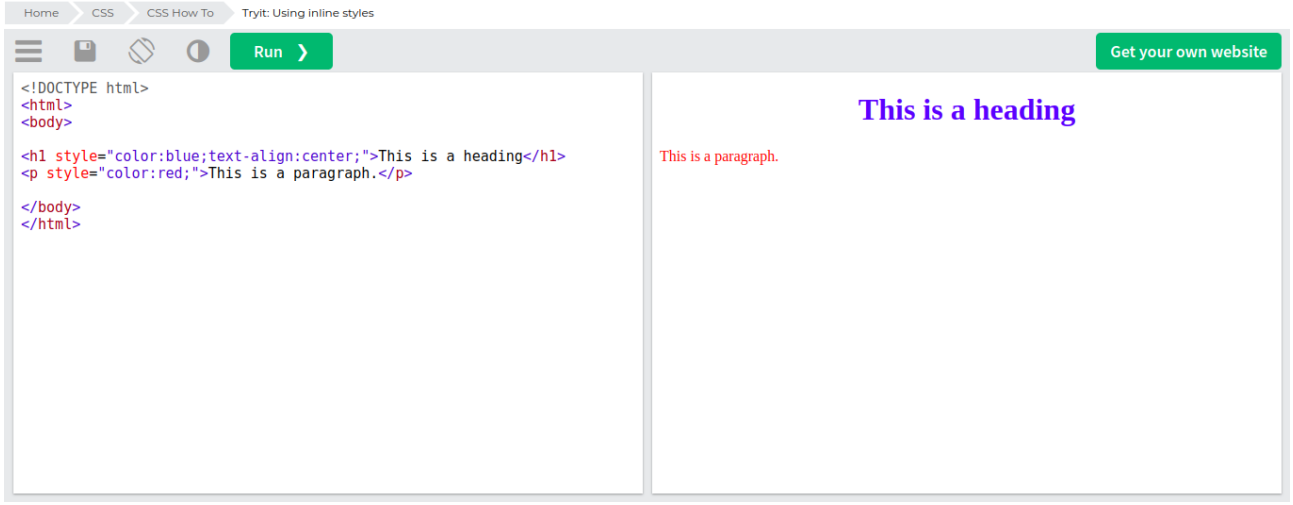


3.Inline CSS

An inline style may be used to apply a unique style for a single element.

To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

SAMPLE CODE AND OUTPUT:



**** Cascading Order ****

What style will be used when there is more than one style specified for an HTML element?

All the styles in a page will "cascade" into a new "virtual" style sheet by the following rules, where number one has the highest priority:

1. Inline style (inside an HTML element)
2. External and internal style sheets (in the head section)
3. Browser default

So, an inline style has the highest priority, and will override external and internal styles and browser defaults.

TRY THIS:

Apply different levels of styles to a single element and observe the difference?

CSS Comments:

Comments are used to explain the code, and may help when you edit the source code at a later date.

Comments are ignored by browsers.

A CSS comment is placed inside the `<style>` element, and starts with `/*` and ends with `*/`

SAMPLE CODE:

```
* This is a  comment *
p {
  color: red;
}
```

Overview of html and css comments:

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red; /* Set text color to red */
}
</style>
</head>
<body>

<h2>My Heading</h2>

<!-- These paragraphs will be red -->
<p>Hello World!</p>
<p>This paragraph is styled with CSS.</p>
<p>CSS comments are not shown in the output.</p>

</body>
</html>
```

CSS COLORS:

.Colors are specified using predefined color names, or RGB, HEX, HSL, RGBA, HSLA values.

Using color Names:

> You are supposed to specify the color names directly.
Ex: orange,DodgerBlue,pink,white,black etc;

Using RGB values:

>In CSS, a color can be specified as an RGB value, using this formula:

rgb(red, green, blue)

>Each parameter (red, green, and blue) defines the intensity of the color between 0 and 255.

>For example, rgb(255, 0, 0) is displayed as red, because red is set to its highest value (255) and the others are set to 0.

>To display black, set all color parameters to 0, like this: rgb(0, 0, 0).

>To display white, set all color parameters to 255, like this: rgb(255, 255, 255).

Using HEX Values:

>In CSS, a color can be specified using a hexadecimal value in the form:

#rrggbb

>Where rr (red), gg (green) and bb (blue) are hexadecimal values between 00 and ff (same as decimal 0-255).

>For example, #ff0000 is displayed as red, because red is set to its highest value (ff) and the others are set to the lowest value (00).

>To display black, set all values to 00, like this: #000000.

>To display white, set all values to ff, like this: #ffffff.

SAMPLE CODE:

```
<!-- color Names -->
<h1 style="background-color:DodgerBlue;">Hello World</h1>
<p style="background-color:Tomato;">Lorem ipsum...</p>
<!-- RGB values: -->
```

```
<h1 style="background-color:rgb(255,230,2);">Hello World</h1>
<p style="background-color:rgb(233,23,33);">Lorem ipsum...</p>

<!-- HEX Values: -->
<h1 style="background-color:#ff0000;">Hello World</h1>
<p style="background-color:#00ff00;">Lorem ipsum...</p>
```

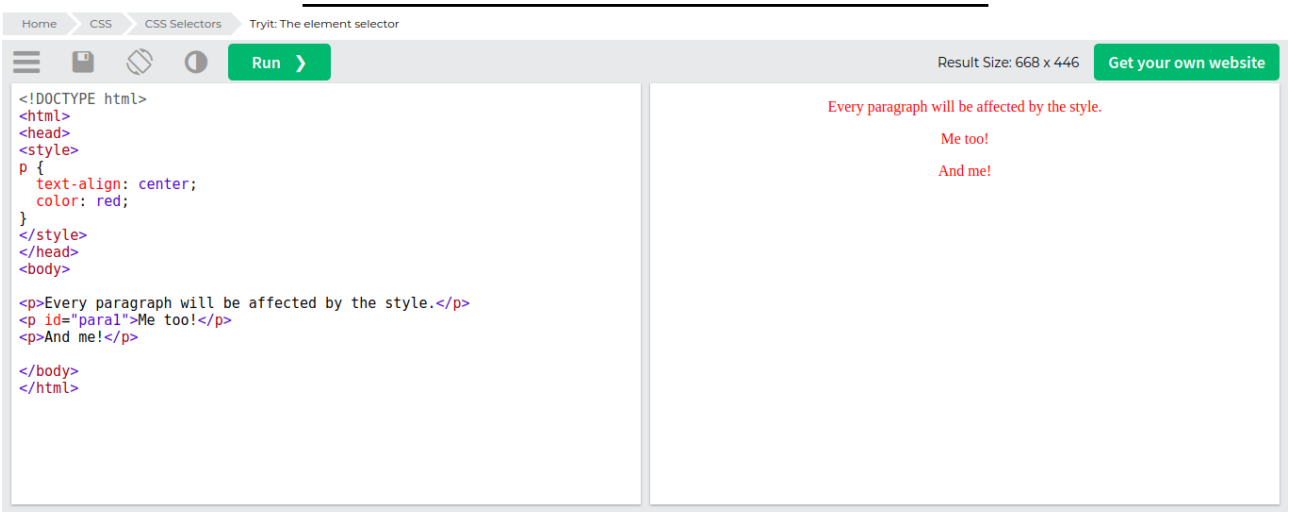
CSS Selectors

CSS selectors are used to "find" (or select) the HTML elements you want to style.

1.The CSS element Selector

The element selector selects HTML elements based on the element name.

SAMPLE CODE AND OUTPUT:



2.The CSS id Selector

The id selector uses the id attribute of an HTML element to select a specific element.

The id of an element is unique within a page, so the id selector is used to select one unique element!

To select an element with a specific id, write a hash (#) character, followed by the id of the element.

SAMPLE CODE AND OUTPUT:



> NOTE:An id name cannot start with a number!

3.The CSS class Selector

The class selector selects HTML elements with a specific class attribute.

To select elements with a specific class, write a period (.) character, followed by the class name.

SAMPLE CODE AND OUTPUT:



NOTE: An id name cannot start with a number!

4.The CSS Universal Selector

The universal selector (*) selects all HTML elements on the page.

SAMPLE CODE AND OUTPUT:

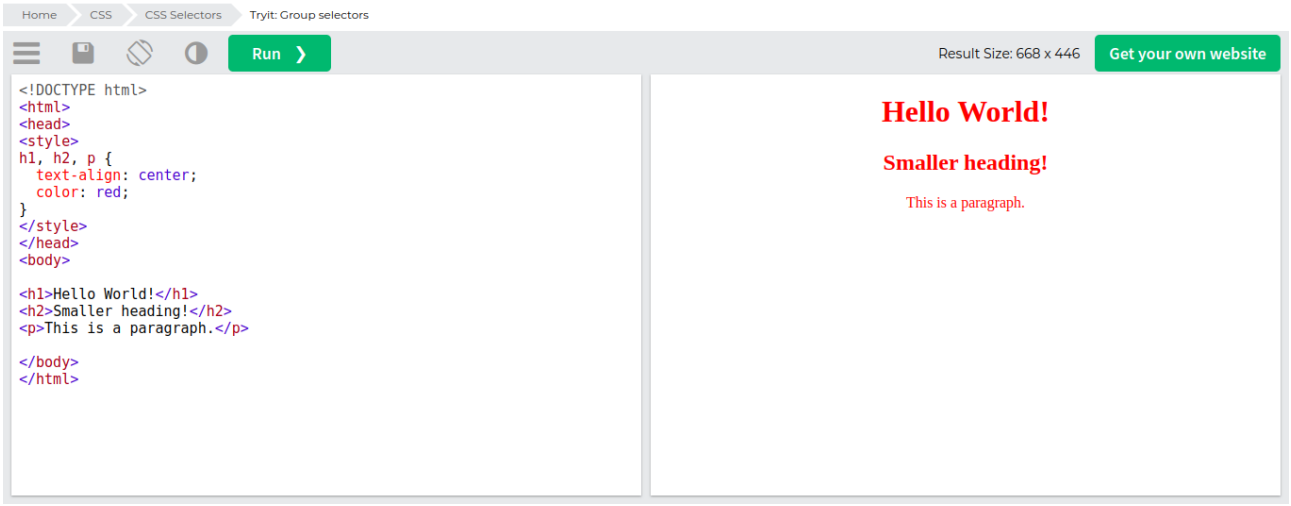


5.The CSS Grouping Selector

The grouping selector selects all the HTML elements with the same style definitions.

i.e.When there are multiple elements having same style definitions ,then we can use grouping selecLetor.

SAMPLE CODE AND OUTPUT:



CSS Backgrounds:

The CSS background properties are used to add background effects for elements.

1.CSS background-color

The `background-color` property specifies the background color of an element.

SAMPLE CODE:

```
body {
  background-color: lightblue;
}
```

2.CSS background-image

The `background-image` property specifies an image to use as the background of an element.

By default, the image is repeated so it covers the entire element.

SAMPLE CODE:

```
body {
  background-image: url("paper.gif");
```

3.CSS background-repeat

By default, the `background-image` property repeats an image both horizontally and vertically.

Some images should be repeated only horizontally or vertically,

SAMPLE CODE:

```
body {
  background-image: url("gradient_bg.png");
  background-repeat: repeat-x;
}
```

Note:

Other values for background-repeat:

1.repeat-y(image repeated vertically)

2.no-repeat(image can’t be repeated)

4.CSS background-position

The `background-position` property is used to specify the position of the background image.

SAMPLE CODE:

```
body {
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
}
```

Note:

Other Values of background-position:

left top
left center
left bottom
right top
right center
right bottom

center top
center center
center bottom

5.CSS background-attachment

The `background-attachment` property specifies whether the background image should scroll or be fixed (will not scroll with the rest of the page):

SAMPLE CODE:

```
body {
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
  background-attachment: fixed;
}

/* scroll - this value enables the image scrolling with the rest of page. */
```



***CSS background - Shorthand property

To shorten the code, it is also possible to specify all the background properties in one single property. This is called a shorthand property.

Instead of writing:

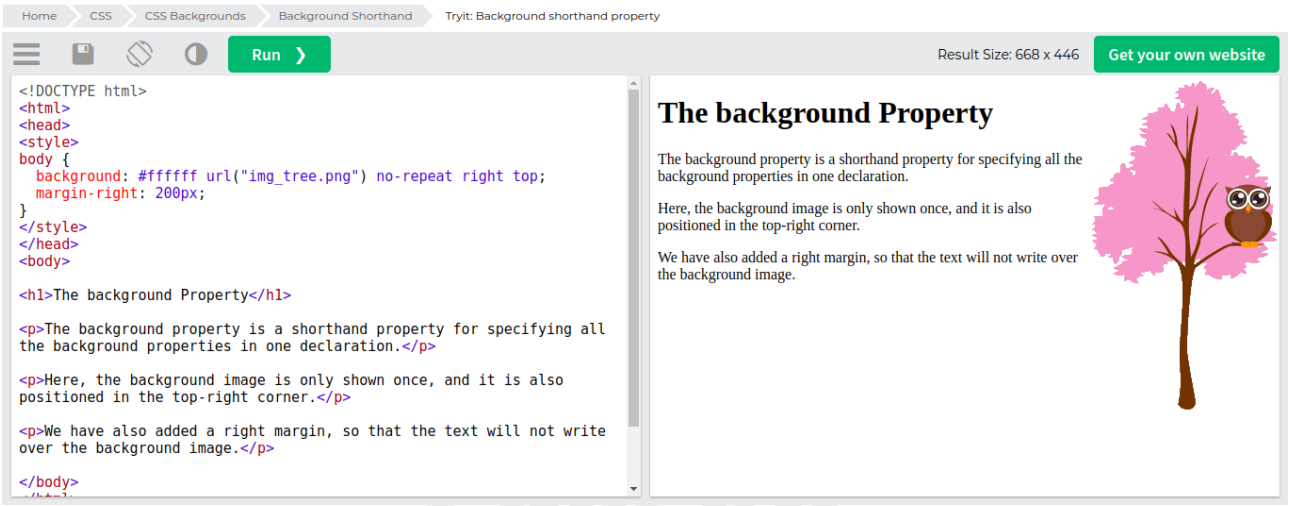
```
body {
  background-color: #ffffff;
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
```

```
  background-attachment:fixed;
  background-position: right top;
}
```

SHORTHAND Propery:

```
body {
  background: #ffffff url("img_tree.png") no-repeat right
top;
}
/*
background: #color #image #repeat #attachment #postion;
*/
```

SAMPLE CODE AND OUTPUT:



CSS Borders

The CSS border properties allow you to specify the style, width, and color of an element's border.

1.CSS Border Style

The `border-style` property specifies what kind of border to display.

The following values are allowed:

- `dotted` - Defines a dotted border
- `dashed` - Defines a dashed border
- `solid` - Defines a solid border
- `double` - Defines a double border
- `groove` - Defines a 3D grooved border. The effect depends on the border-color value
- `ridge` - Defines a 3D ridged border. The effect depends on the border-color value
- `inset` - Defines a 3D inset border. The effect depends on the border-color value
- `outset` - Defines a 3D outset border. The effect depends on the border-color value
- `none` - Defines no border
- `hidden` - Defines a hidden border

SAMPLE CODE:

```
p{
    border-style:dotted;
}
```

2.CSS Border Width

The `border-width` property specifies the width of the four borders.

The width can be set as a specific size (in px, pt, cm, em, etc) or by using one of the three pre-defined values: thin, medium, or thick:

SAMPLE CODE:

p{

border-width:10px;/* all four sides have width 10px*/

}

```
p{
  border-style: solid;
  border-width: 5px 20px; /* 5px top and bottom, 20px on the sides */
}

p{
  border-style: solid;
  border-width: 20px 5px; /* 20px top and bottom, 5px on the sides */
}

p{
  border-style: solid;
  border-width: 25px 10px 4px 35px; /* 25px top, 10px
```

```
right, 4px bottom and 35px left */
}
```

3.CSS Border Color

The `border-color` property is used to set the color of the four borders.

SAMPLE CODE:

```
p{
  border-style: solid;
  border-color: green;
}

p{
  border-style: dotted;
  border-color: blue;
}
```

Note:

**CSS Border Sides:

If the `border-style` property has four values:

- **border-style: dotted solid double dashed;**
  - top border is dotted
  - right border is solid
  - bottom border is double
  - left border is dashed

If the `border-style` property has three values:

- **border-style: dotted solid double;**

- top border is dotted
- right and left borders are solid
- bottom border is double

If the `border-style` property has two values:

- **border-style: dotted solid;**
  - top and bottom borders are dotted
  - right and left borders are solid

If the `border-style` property has one value:

- **border-style: dotted;**
  - all four borders are dotted



***CSS Border - Shorthand Property

To shorten the code, it is also possible to specify all the individual border properties in one property.

The `border` property is a shorthand property for the following individual border properties:

- `border-width`
- `border-style` (required)
- `border-color`

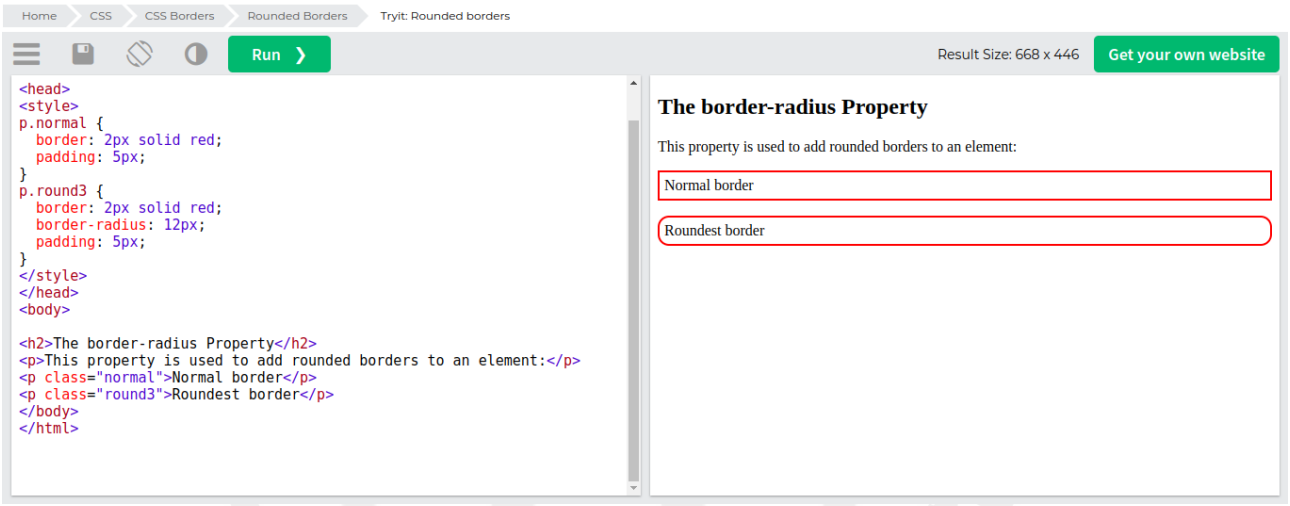
SAMPLE CODE AND OUTPUT:



***** CSS Rounded Borders*****

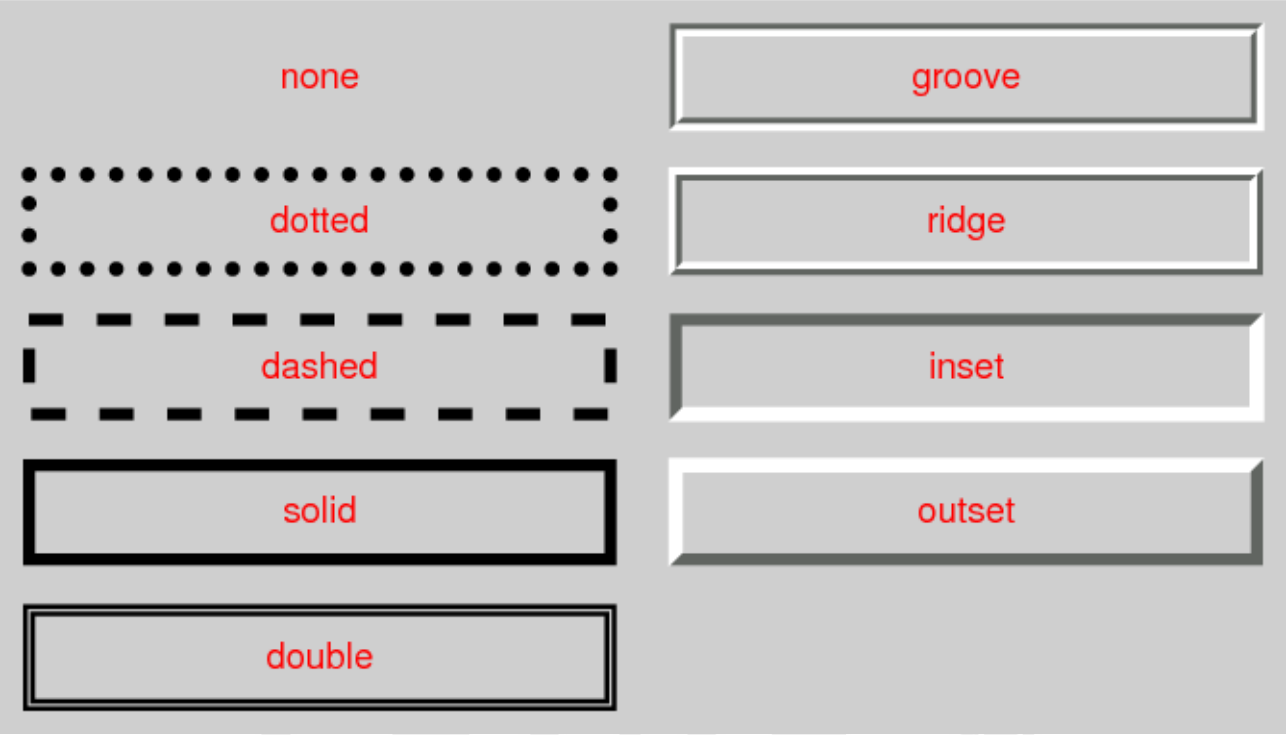
The `border-radius` property is used to add rounded borders to an element:

SAMPLE CODE AND OUTPUT:



Discuss: What happens radius is set to 50% when height is equal to widht of any HTML Element?

Have a look



CSS Margins

The CSS `margin` properties are used to create space around elements, outside of any defined borders.

With CSS, you have full control over the margins. There are properties for setting the margin for each side of an element (top, right, bottom, and left).

CSS has properties for specifying the margin for each side of an element:

- `margin-top`
- `margin-right`
- `margin-bottom`
- `margin-left`

All the margin properties can have the following values:

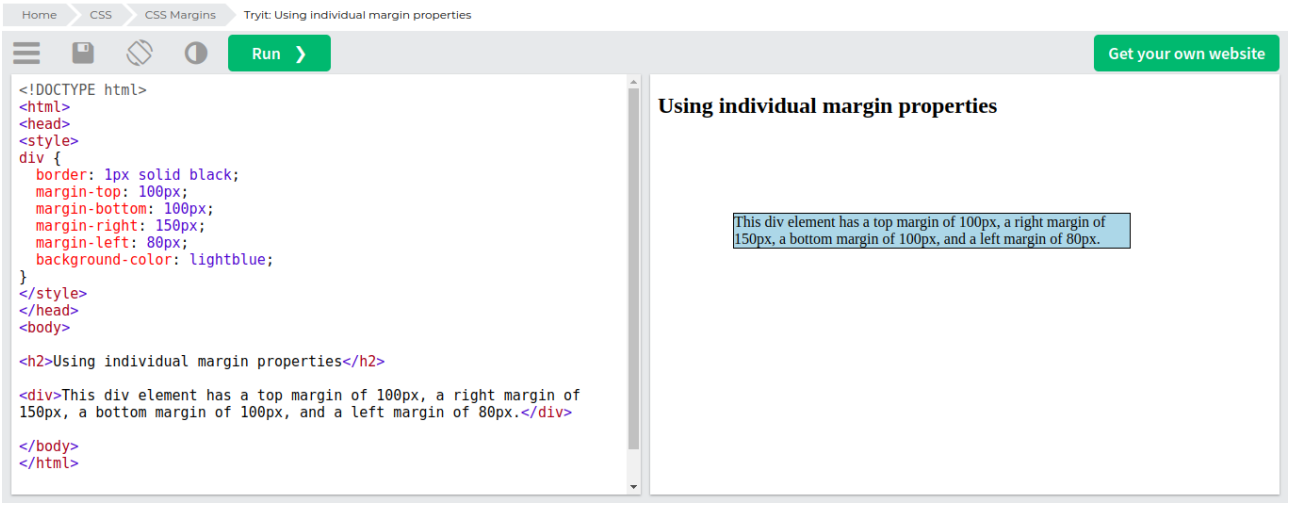
- auto - the browser calculates the margin
- length - specifies a margin in px, pt, cm, etc.
- % - specifies a margin in % of the width of the containing element
- inherit - specifies that the margin should be inherited from the parent element

Tip: Negative values are allowed.

SAMPLE CODE:

```
/* It will be applied to all four sides around the element
*/
p {
  margin:20px;
}
/* you can specify for single side */
p {
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
}
```

LOOK this for better understanding:



DISCUSS:what happens for below code?;

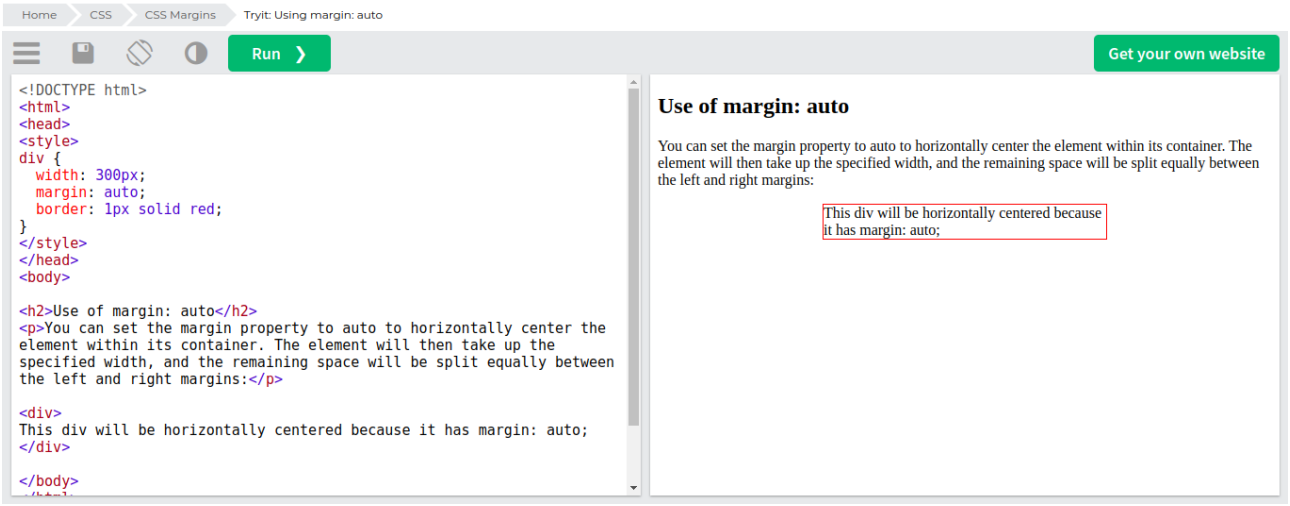
```
p{
   margin:20px 20px;
}
```

**The auto Value

You can set the margin property to auto to horizontally center the element within its container.

The element will then take up the specified width, and the remaining space will be split equally between the left and right margins.

SAMPLE CODE:



CSS Padding

Padding is used to create space around an element's content, inside of any defined borders.

CSS has properties for specifying the padding for each side of an element:

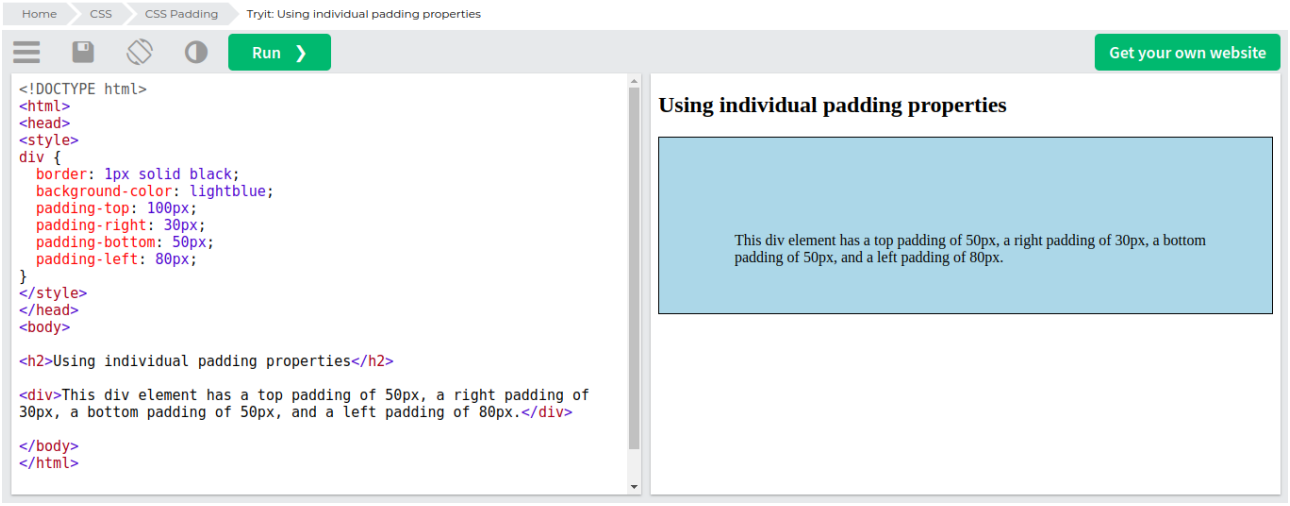
- `padding-top`
- `padding-right`
- `padding-bottom`
- `padding-left`

All the padding properties can have the following values:

- length - specifies a padding in px, pt, cm, etc.
- % - specifies a padding in % of the width of the containing element
- inherit - specifies that the padding should be inherited from the parent element

Note: Negative values are not allowed.

SAMPLE CODE AND OUTPUT:



Note:

```
/* short hand propery
    padding: #top #right #bottom #left;
    It is similar to margin property
*/
```

CSS HEIGHT/WIDTH

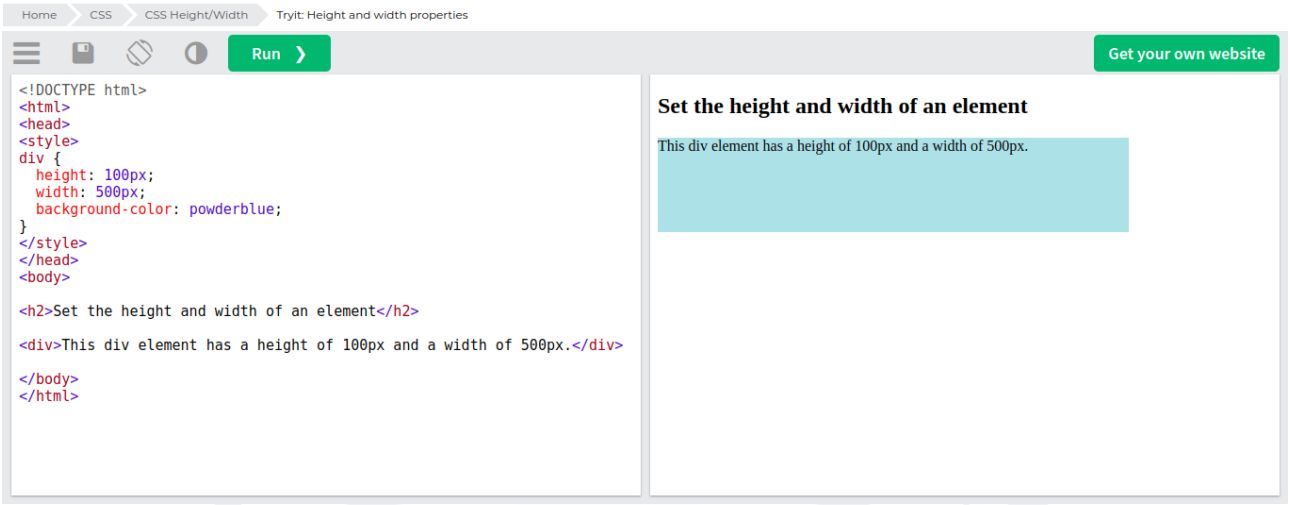
The `height` and `width` properties are used to set the height and width of an element.

The height and width properties do not include padding, borders, or margins. It sets the height/width of the area inside the padding, border, and margin of the element.

The `height` and `width` properties may have the following values:

- `auto` - This is default. The browser calculates the height and width
- `length` - Defines the height/width in px, cm, etc.
- `%` - Defines the height/width in percent of the containing block
- `initial` - Sets the height/width to its default value
- `inherit` - The height/width will be inherited from its parent value

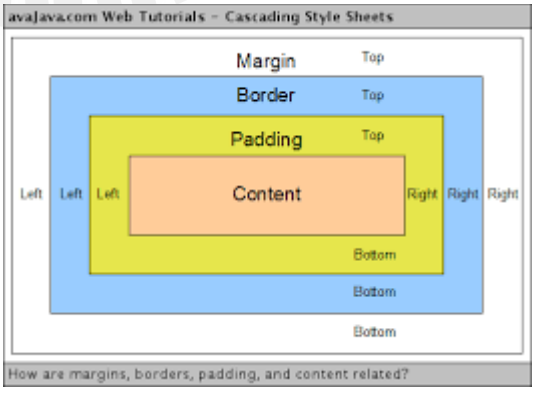
SAMPLE CODE AND OUTPUT:



CSS Box Model

All HTML elements can be considered as boxes.

The CSS box model is essentially a box that wraps around every HTML element. It consists of: content,

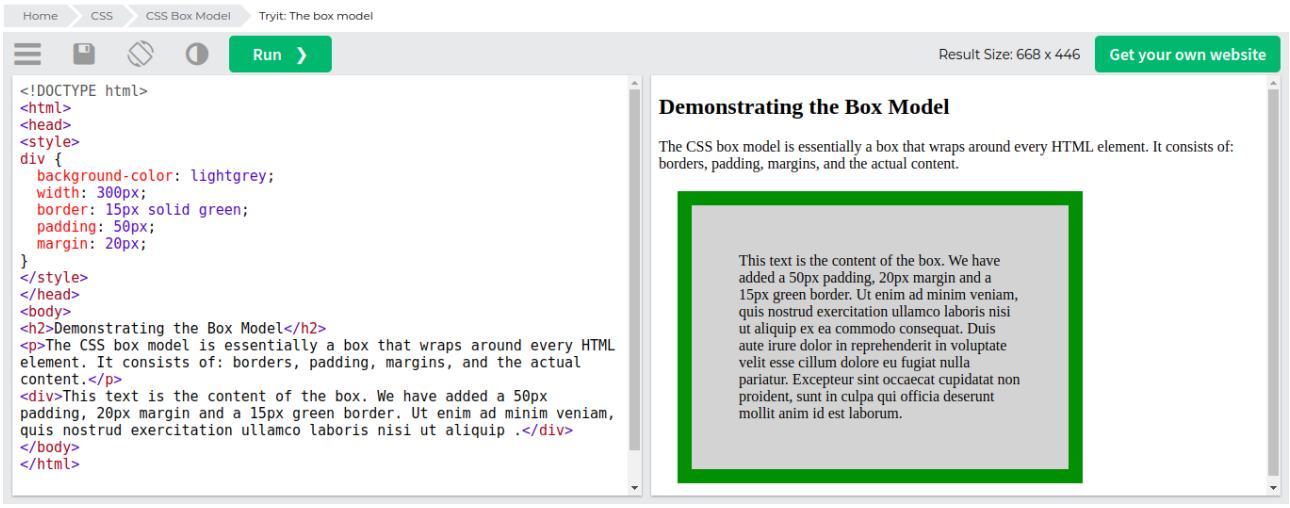


padding, borders and margins. The image above illustrates the box model:

Explanation of the different parts:

- **Content** - The content of the box, where text and images appear
- **Padding** - Clears an area around the content. The padding is transparent
- **Border** - A border that goes around the padding and content
- **Margin** - Clears an area outside the border. The margin is transparent

SAMPLE CODE AND OUTPUT:



For daily Challenge:

Refer **IcodeThis webstite .** It provides daily challenging template that can be built by using html and css as well as javascript.

CSS Box Sizing

The CSS `box-sizing` property allows us to include the padding and border in an element's total width and height.

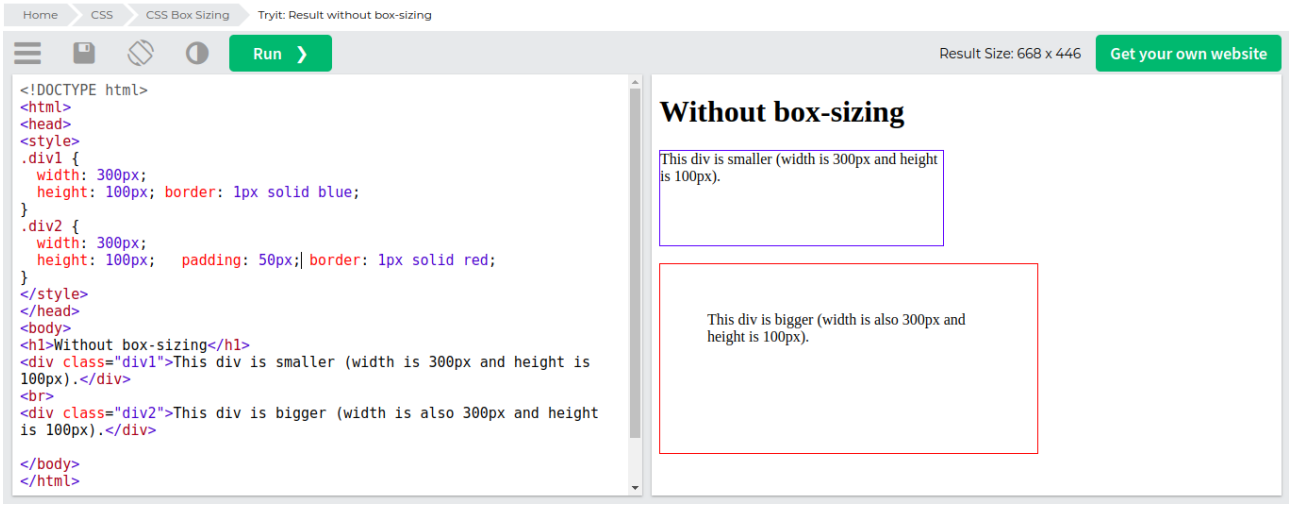
Without the CSS box-sizing Property

By default, the width and height of an element is calculated like this:

width + padding + border = actual width of an element
height + padding + border = actual height of an element

This means: When you set the width/height of an element, the element often appears bigger than you have set (because the element's border and padding are added to the element's specified width/height).

SAMPLE CODE AND OUTPUT:

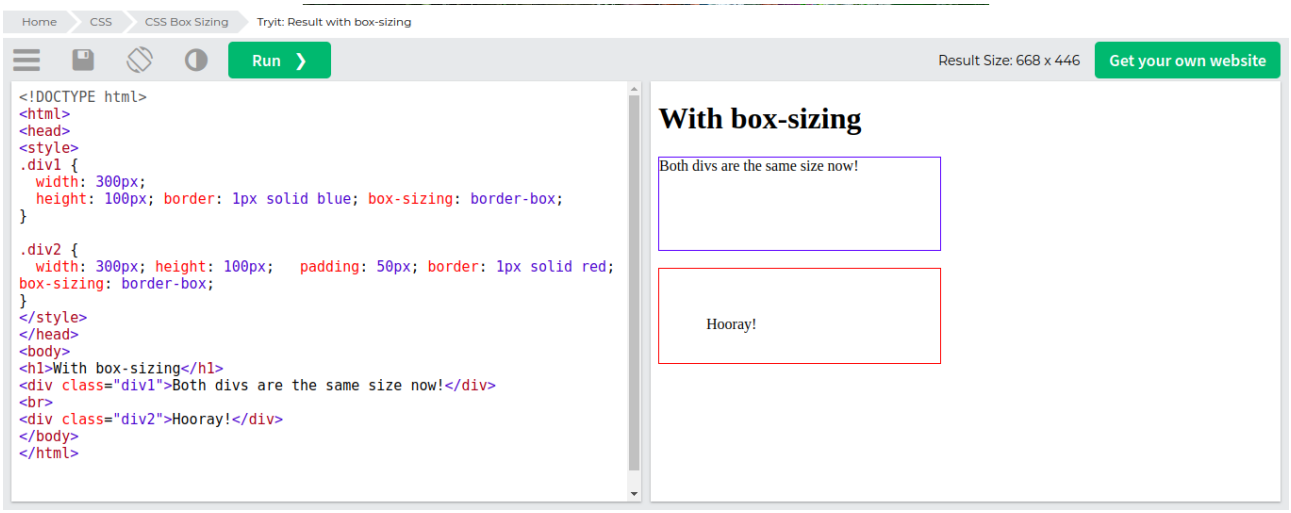


With the CSS box-sizing Property

The `box-sizing` property allows us to include the padding and border in an element's total width and height.

If you set `box-sizing: border-box;` on an element, padding and border are included in the width and height:

SAMPLE CODE AND OUTPUT:



CSS Text

CSS has a lot of properties for formatting text.

1.Text Color:

The `color` property is used to set the color of the text. The color is specified by:

- a color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

2.CSS Text Alignment:

The `text-align` property is used to set the horizontal alignment of a text.

A text can be left or right aligned, centered, or justified.

3.Text Direction:

The `direction` and `unicode-bidi` properties can be used to change the text direction of an element:

4.Text Decoration:

The `text-decoration-line` property is used to add a decoration line to text.

Values:underline,overline,line-through

The `text-decoration-color` property is used to set the color of the decoration line.

Values: any color

The `text-decoration-style` property is used to set the style of the decoration line.

Values: dotted,wavy,double,dashed

The `text-decoration-thickness` property is used to set the thickness of the decoration line.

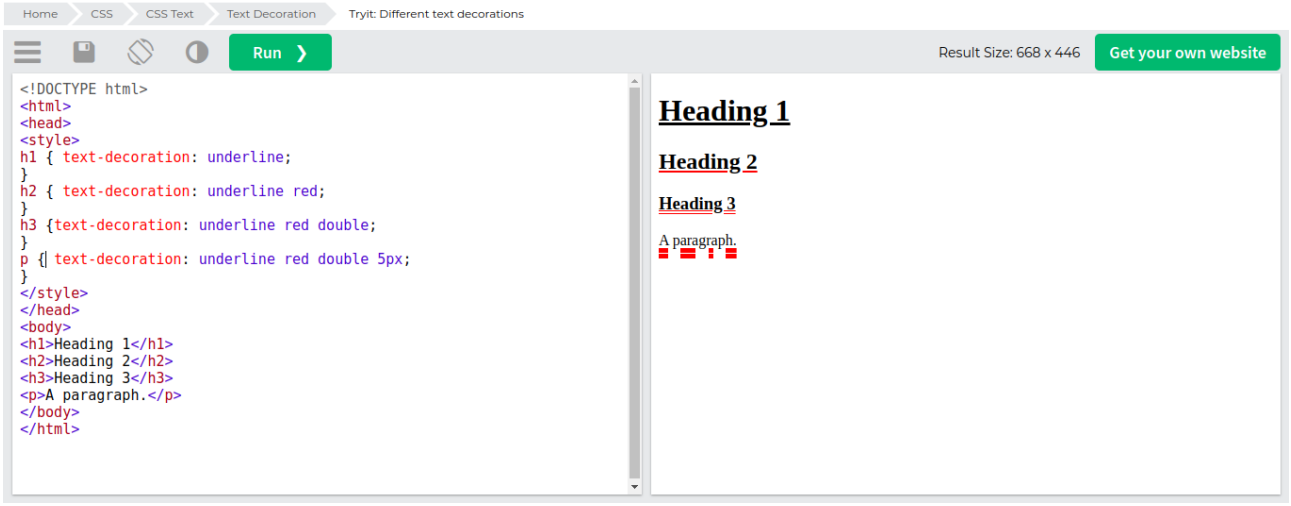
Values:you can specify any thing px,%

**** you can specify each individual property whenever needed or otherwise use shorthand property below mentioned****

SYNTAX(for shorthand):

text-decoration:#line #color #style #thickness

SAMPLE CODE AND OUTPUT:



5.Text Transformation:

The `text-transform` property is used to specify uppercase and lowercase letters in a text.
It can be used to turn everything into uppercase or lowercase letters, or capitalize the first letter of each word:

sample code:

```
    p.uppercase {
  text-transform: uppercase;
}

p.lowercase {
  text-transform: lowercase;
}

p.capitalize {
  text-transform: capitalize;
}
```

6.Text Spacing:

>Text Indentation

The `text-indent` property is used to specify the indentation of the first line of a text:

>Letter Spacing

The `letter-spacing` property is used to specify the space between the characters in a text.

>Line Height

The `line-height` property is used to specify the space between lines

>Word Spacing

The `word-spacing` property is used to specify the space between the words in a text.

>Text Shadow

The `text-shadow` property adds shadow to text.

In its simplest use, you only specify the horizontal shadow (2px) and the vertical shadow (2px):

Syntax:

text-shadow: #horizontal_shadow #vertical_shadow #blur #color;

Ex:

text-shadow:2px 2px 5px red;

SAMPLE CODE AND OUTPUT:

Home > CSS > CSS Text > Text Spacing > Tryit: Using word-spacing

Run ❯ | Result Size: 668 x 446 | Get your own website

```
<!DOCTYPE html>
<html>
<head><!--note you can also specify negative lengths-->
<style>
p.prasad{
    line-height:36px;}
p.one {
  text-indent:20px; word-spacing: 10px;
}

p.two {
  letter-spacing:2px; /*word-spacing: -2px;*/
}
</style>
</head>
<body>
<h1>Using word-spacing</h1>
<p  class="prasad">This is a paragraph with normal word spacing.HI all
I am kancharla Prasad.Have a nice day .Enjoy the day by doing coding.
</p>
<p class="one">This is a paragraph with larger word spacing.</p>
<p class="two">This is a paragraph with smaller word spacing.</p>
</body>
</html>
```

Using word-spacing

This is a paragraph with normal word spacing.HI all I am kancharla Prasad.Have a nice day .Enjoy the day by doing coding.

This is a paragraph with larger word spacing.

This is a paragraph with smaller word spacing.

For Text Shadow:

Home > CSS > CSS Text > Text Shadow > Tryit: Text shadow on white text

Run ❯ | Result Size: 668 x 446 | Get your own website

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  margin-top:100px;
  text-align:center;
  color: red;
  text-shadow: 2px 2px 0px blue;
}
h2{
    margin-top:150px;
    font-size:30px;
    text-align:center;
    transform:rotate(90deg);
    color:red;
    text-shadow:-20px -20px 0px blue;
}
</style>
</head>
<body>

<h1>Programming Club</h1>
<h2> Web Development</h2>
```

Programming Club

Web Development
Web Development

I hope everyone enjoyed a lot today

```
Have
    a
    Nice
        Day
```